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Radio and the Rise of the Nazis in Prewar Germany

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1 Big picture

- **Question.** How did radio affect political support for democratic institutions and for the Nazi regime in Germany during the collapse of the Weimar Republic and the consolidation of dictatorship?
- **Setting.** The paper studies Germany from the late 1920s through the 1930s. Radio expanded rapidly at exactly the time when radio content changed from mostly apolitical programming, to pro-Weimar political programming, and then to pro-Nazi propaganda after Hitler became chancellor.
- **Core empirical idea.** The authors combine two forms of variation: changes over time in who controlled radio content, and cross-sectional plus time-series variation in access to radio generated by transmitter location, transmitter power, terrain, and radio-signal propagation.
- **Main content regimes.** Before 1929, radio was largely apolitical. From 1929 to January 1933, radio carried progovernment and anti-extremist political content, while the Nazis and communists were mostly denied airtime. After January 30, 1933, radio shifted sharply toward pro-Nazi propaganda.
- **Main result during democracy.** When radio was pro-Weimar, radio exposure slowed the growth of Nazi vote share. When the Nazis gained control of radio during the campaign for the March 1933 election, radio exposure increased Nazi vote share.
- **Counterfactual magnitudes.** In the absence of radio in the September 1930 election, the Nazis are estimated to have received about 22.3% of the vote instead of 18.2%. In March 1933, without Nazi radio access, the Nazi vote share is estimated to have been about 41.0% instead of 43.9%.
- **Main result after dictatorship consolidated.** Nazi radio propaganda increased anti-Semitic behavior by ordinary Germans: deportations based partly on denunciations and open anti-Semitic letters to *Der Stuermer* rose in areas with stronger radio signal.
- **Predisposition mechanism.** Propaganda was strongest where the audience was historically predisposed toward the message: places with medieval anti-Jewish pogroms, stronger extreme-nationalist voting in 1924, or high historical land inequality. In places with Jewish settlement but no medieval pogroms, Nazi radio appears to have backfired.
- **Interpretation.** Media is not inherently democratic or authoritarian. Its effect depends on who controls content and whether listeners are receptive to the message.
- **Takeaway.** Radio could protect an unconsolidated democracy when controlled by prodemocratic forces, but the same technology became a powerful tool of authoritarian consolidation once extremists captured the state.

2 Data and measurement (key objects)

2.1 Historical timing and radio-content regimes

The paper divides German radio history before World War II into three main regimes.

- **Apolitical radio before 1929.** Programming mainly consisted of music, literature, lectures, weather, sports, and other cultural content. Election campaigning was largely absent from radio.
- **Pro-Weimar radio, 1929–January 1933.** After the right-wing campaign for a referendum against the Young Plan, the Weimar government introduced political news and commentary. The content had a progovernment and anti-extremist slant. The Nazis were mostly denied radio access.
- **Pro-Nazi radio after January 30, 1933.** After Hitler was appointed chancellor, the Nazis gained control of radio. During the five-week campaign before the March 1933 election, Hitler and Nazi officials appeared repeatedly on the air. Later, radio became a central tool of Nazi propaganda.

Key variable. The paper summarizes the political direction of content with a slant variable:

$$\text{Slant}_t = \begin{cases} 0, & \text{before political radio became important, especially 1928,} \\ -1, & \text{pro-Weimar / anti-extremist radio, 1929–1932,} \\ +1, & \text{pro-Nazi radio, 1933 onward.} \end{cases} \quad (1)$$

2.2 Radio exposure

The paper uses three related measures of exposure.

- **Radio signal strength.** The authors compute predicted signal strength using transmitter location, frequency, power, terrain, and the irregular terrain model. This measure is available for districts and cities across the full period.
- **Nonlinear transformation of signal strength.** Signal strength affects listening in an S-shaped way. Very weak signals do not allow reliable listening; once signal is already strong enough, extra signal has little marginal effect. The nonlinear transformation is chosen to fit actual subscription rates.
- **Radio subscription rate.** Subscription data are available at the district level for April 1931, April 1932, and April 1933. The paper uses these data directly when possible, and also uses predicted subscriptions based on signal strength.

Measurement interpretation. Signal strength is not identical to listening. It is an instrument-like proxy for potential access. Subscription rates are closer to actual listening but are available at fewer dates and may be endogenous to local socioeconomic conditions.

2.3 Political outcomes during the democratic period

The main democratic-period outcomes are district-level electoral outcomes.

- **Parliamentary elections.** Nazi vote shares are observed for the Reichstag elections of 1928,

September 1930, July 1932, November 1932, and March 1933.

- **Presidential election.** The paper studies the 1932 presidential election, especially votes for incumbent President von Hindenburg and for Hitler.
- **Referendum.** The paper studies the 1929 referendum against the Young Plan. The outcome is the share of eligible voters voting yes, because turnout was so low that abstention effectively worked like voting no.
- **Turnout and other parties.** These are used in robustness checks and as controls.

Important election numbers:

- Nazi vote share rose from 2.6% in 1928 to 18.3% in September 1930.
- It reached 37.3% in July 1932 and fell to 33.1% in November 1932.
- It rose to 43.9% in March 1933, the last semi-competitive election.

2.4 Anti-Semitism and party membership outcomes

The paper also studies non-electoral outcomes that measure political behavior and support for Nazi policies.

- **Discrimination against Jews, 1929–1934.** A city-year panel records verbal anti-Semitic expressions, physical violence, and property damage against Jews.
- **New NSDAP membership.** The paper uses party-membership-card data to measure new Nazi party members in 1932 and in February–May 1933.
- **Deportations before 1942.** Before deportations became fully systematic, they partly reflected local denunciations and local hostility. The paper uses the number of deported Jews as an outcome.
- **Letters to *Der Stürmer*.** Anti-Semitic letters to the Nazi newspaper from 1935 to 1938 are used as a measure of open anti-Semitic expression.
- **Kristallnacht synagogue attacks.** The paper uses whether synagogues or Jewish prayer rooms were damaged or destroyed in 1938, though this outcome is less strongly related to radio because attacks were heavily organized by party officials.

2.5 Predisposition to propaganda

A central feature of the paper is that propaganda effects depend on the audience's prior beliefs and social environment.

- **Medieval pogroms.** The occurrence of anti-Jewish pogroms during the Black Death, 1348–1350, proxies for persistent local anti-Semitism.
- **Jewish settlement without pogroms.** Places with historical Jewish settlement but no pogroms are interpreted as less predisposed toward anti-Semitic messages.

- **Extreme-right voting in 1924.** Vote share for the National Socialist Freedom Party, when the NSDAP was banned, proxies for preexisting extreme-nationalist support.
- **Historical land inequality.** Landholding inequality in 1895 proxies for social conflict and susceptibility to extremist messages during economic crisis.

2.6 Controls

The baseline controls include geography, socioeconomic structure, and preexisting political preferences.

- Geographic controls: altitude, city status, river location, and distance to the nearest large city.
- Socioeconomic controls: population, Jewish and Catholic shares, blue-collar and white-collar shares, unemployment, welfare recipients, social-housing renters, war participants, and property-tax measures.
- Political controls: 1924 turnout and vote shares for DNVP, NSFP, SPD, and Zentrum.
- Fixed effects: region fixed effects, time fixed effects, and district or city fixed effects in panel specifications.

3 Models and empirical specifications

3.1 Panel specification

The main panel specification is:

$$y_{it} = \beta_0 + \beta_1 \text{Slant}_t \times \text{Exposure}_{it} + Z'_{it} \beta_2 + \phi_i + \lambda_t + \varepsilon_{it}. \quad (2)$$

Here, y_{it} is an outcome such as Nazi vote share or anti-Semitic incidents. i indexes districts or cities and t indexes election years or calendar years. Exposure_{it} is radio exposure. ϕ_i are district or city fixed effects. λ_t are time fixed effects. Z_{it} includes interactions of time-invariant controls with time fixed effects.

Interpretation of β_1 . The coefficient measures whether radio exposure has different effects depending on the direction of radio slant. Since $\text{Slant}_t = -1$ in the pro-Weimar period and $\text{Slant}_t = +1$ in the pro-Nazi period, the same technology can have opposite political effects.

3.2 First-difference event specifications

To isolate the two major content changes, the paper estimates first differences around two events:

$$\Delta y_{it} = \gamma_0 + \gamma_1(t) \text{Exposure}_{it} + X'_i \theta + \varepsilon_{it}, \quad (3)$$

where $t = 1930$ or $t = 1933$.

- For 1930, $\Delta y_{i,1930}$ is the change in Nazi vote share from 1928 to September 1930. This captures the effect of introducing pro-Weimar political radio.

- For 1933, $\Delta y_{i,1933}$ is the change in Nazi vote share from November 1932 to March 1933. This captures the effect of the switch to pro-Nazi radio after Hitler became chancellor.

Expected signs. The 1930 coefficient should be negative because radio was anti-extremist and pro-Weimar. The 1933 coefficient should be positive because the Nazis had captured radio content.

3.3 Cross-sectional specifications

For outcomes without a long panel, the paper estimates:

$$k_i = \delta_0 + \delta_1 \text{Exposure}_i + X_i' \alpha_2 + \varepsilon_i. \quad (4)$$

This specification is used for outcomes such as Nazi party membership, referendum and presidential-election outcomes, deportations, anti-Semitic letters, and synagogue attacks.

Interpretation. The coefficient δ_1 compares places with higher and lower radio exposure, conditional on baseline controls and region fixed effects. This is more demanding than the panel specification because it relies more heavily on cross-sectional comparability.

3.4 Heterogeneity by predisposition

The propaganda-heterogeneity specification is:

$$k_i = \delta_0 + \delta_1 \text{Exposure}_i \quad (5)$$

$$+ \delta_2 (\text{Exposure}_i - \overline{\text{Exposure}}) P_i + \delta_3 P_i + X_i' \alpha_4 + \varepsilon_i. \quad (6)$$

P_i is a measure of predisposition to propaganda, such as medieval pogroms, extreme-nationalist voting, or land inequality.

Interpretation. The coefficient δ_2 asks whether radio propaganda is more effective where listeners are more receptive. A positive δ_2 means propaganda works more strongly in places predisposed toward its message.

3.5 Instrumental-variables logic

When actual subscriptions are the exposure measure, the paper instruments subscription rates with predicted subscriptions based on signal strength.

$$\text{Subscription}_{it} = \pi_0 + \pi_1 \text{Pred}(\text{Signal}_{it}) + X_i' \pi_2 + u_{it}. \quad (7)$$

The second stage replaces radio exposure with predicted exposure from the first stage. The exclusion idea is that signal strength affects political outcomes mainly through radio listening, after controlling for geography, socioeconomic characteristics, region effects, and preexisting voting.

Important nuance. Signal strength itself is not randomly assigned. Transmitters were placed to reach people. The paper therefore relies on detailed controls, fixed effects, placebo tests, and Altonji–Elder–Taber style diagnostics.

4 Identification and credibility

4.1 What identifies the effects?

The credibility of the design comes from the interaction of three facts.

- Radio access varied across space because of transmitter placement, transmitter upgrades, distance, and terrain.
- Radio access varied over time because the radio network expanded during the 1920s and 1930s.
- The political slant of radio changed sharply over time: apolitical, then pro-Weimar, then pro-Nazi.

The core identification question is not simply whether radio places differed from nonradio places. The paper asks whether places with more radio exposure changed differently exactly when the content of radio changed.

4.2 Why transmitter placement is a concern

Transmitters were placed strategically, not randomly. Signal strength is correlated with population density, cities, income, and political conditions. Therefore, a naive cross-sectional comparison would be hard to interpret.

The paper addresses this by controlling for:

- region fixed effects;
- distance to large cities, city dummies, altitude, and population polynomials;
- socioeconomic variables;
- early political preferences, especially 1924 votes for major parties and extremist-nationalist parties;
- district or city fixed effects in panel specifications.

4.3 Placebo tests

The paper checks whether future radio availability predicts outcomes measured before radio should matter.

- Radio signal in 1928–1933 does not systematically predict election outcomes from 1920, 1924, 1925, or 1928.
- Future changes in signal strength do not predict past changes in voting outcomes.
- City-level signal strength does not predict pre-radio crime rates or anti-Jewish pogroms from the 1920s.

Economic message. These placebo tests reduce the concern that radio signal is merely proxying for preexisting trends toward Nazism or anti-Semitism.

4.4 Altonji–Elder–Taber style tests

The paper uses AET-style diagnostics. The idea is to construct an index of observables that best predicts signal strength and then test whether that index predicts the outcome in the same way as signal strength.

Logic. If observables that predict radio availability also strongly predict the outcome, unobservables correlated with radio may be a serious concern. If the observable index does not predict the outcome, omitted-variable bias is less plausible.

Across the main specifications, these tests generally do not show a pattern consistent with the observable determinants of radio driving the results.

4.5 Mechanism: persuasion versus coordination

Radio propaganda could matter for at least two reasons.

- **Persuasion.** Listeners update their beliefs and become more supportive of Nazi positions.
- **Coordination.** Radio informs listeners that the regime supports certain actions, reducing fear of punishment or increasing perceived social approval.

The backfiring result in historically less anti-Semitic places is difficult to explain with pure coordination. If radio merely announced that anti-Semitic actions were tolerated, it should not reduce anti-Semitic behavior. The negative effect in some places suggests that persuasion or dissuasion is important.

4.6 Limitations

The paper is unusually rich historically, but some limitations remain.

- Actual listening is imperfectly measured. Subscription rates are only observed at a few dates and only at district level.
- Radio exposure is not randomly assigned. The paper uses many controls and checks, but some residual spatial confounding is always possible.
- Content slant is measured coarsely as direction rather than exact intensity.
- The March 1933 election was not fully free. Nazi radio effects occurred alongside repression, newspaper bans, violence, and the Reichstag Fire Decree.
- Outcomes such as deportations and synagogue attacks reflect a mixture of citizen behavior, party organization, and state coercion.

5 Tables and figures

5.1 Figure I: Timing and political orientation of radio broadcasts

Read. Panel A shows that political broadcasts were rare before 1929 and rose sharply afterward. Panel B shows that before Hitler became chancellor, the Nazis had almost no access to radio, while government officials did. In the March 1933 campaign, Nazi access rose dramatically.

Economic message. The content shock is central. The paper is not estimating a stable effect of radio technology; it estimates how radio worked under different political control regimes.

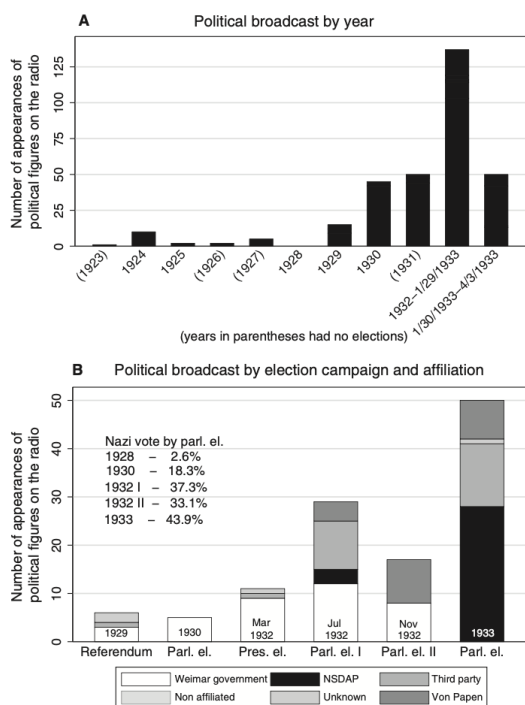


FIGURE I

Timing and Political Orientation of Radio Broadcast

Panel A. The intensity of the political broadcast

Panel B. Access of political figures to radio by election campaign and affiliation

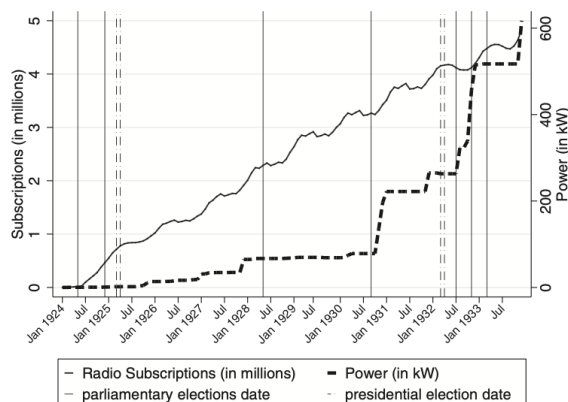


FIGURE II

Number of Radio Subscriptions and Cumulated Power of Transmitters in Germany, 1924-1933

Source: Vaessen (1938) (subscriptions) and the authors' data on transmitters (see Online Appendix: data sources).

5.2 Figure II: Radio subscriptions and transmitter power

Read. Radio subscriptions increased from essentially zero in 1924 to almost 5 million by the end of 1933. Transmitter power also expanded sharply.

Economic message. The timing is useful for identification. Radio became widespread exactly when democratic institutions were fragile and political content was changing.

5.3 Figure III: Timing of outcomes and content changes

Read. The figure aligns radio-content regimes with measured outcomes: elections, referendum voting, Nazi membership, anti-Semitic incidents, deportations, letters to *Der Stuermer*, and Kristallnacht.

Economic message. The empirical design is historical event-study logic: compare outcomes before and after shifts in radio content, using radio exposure as the treatment intensity.

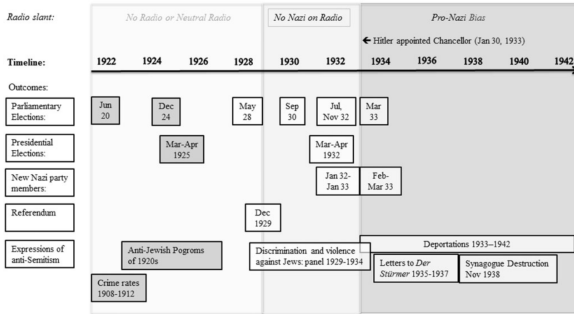


FIGURE III
The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

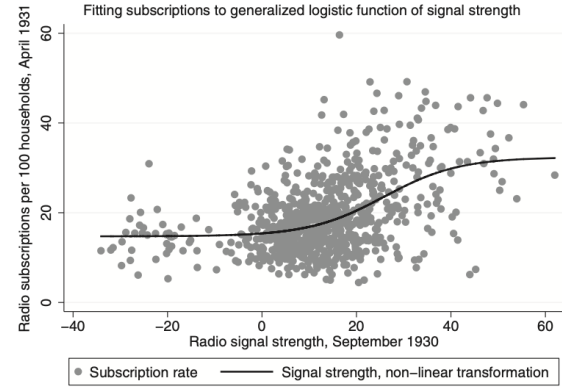


FIGURE IV
Radio Subscriptions and Signal Strength, 1931 (No Controls)
t-statistic for the linear bivariate relationship: 14.12. Source: "Teilnahme am Rundfunk in den einzelnen OPD-Bezirken in Orten mit mehr als 2500 Einwohnern am 1. April 1933," *Veröffentlichungen des Verbandes der Funkindustrie e.V.*, 12, 1933, data for the year 1931.

5.4 Figure IV and Table I: Signal strength and subscription rates

Read. Figure IV shows an S-shaped relationship between signal strength and radio subscription rates. Table I shows that signal strength strongly predicts subscriptions. A one-standard-deviation increase in signal strength is associated with about a 2.8 percentage point increase in subscriptions in 1930 and about a 1.8 percentage point increase in 1933.

Selected Table I coefficients:

- plain signal strength predicts subscription rates with coefficients around 0.16–0.22;
- the nonlinear signal transformation has coefficients around 0.60–0.68;
- first-stage F-statistics are large, supporting instrument relevance.

Economic message. Signal strength is a meaningful predictor of actual radio access, especially after accounting for the nonlinear reception technology.

5.5 Table II: Determinants of radio availability

Read. Radio availability is strongly related to transmitter-location variables: distance to large cities, city status, altitude, and population. Socioeconomic variables and 1924 voting also predict signal strength to some extent.

Economic message. This table is a warning. Radio access is not random. The later empirical design needs fixed effects, controls, placebo tests, and AET-style diagnostics.

5.6 Table III: Radio availability and Nazi vote share, district fixed effects

Read. During September 1930, July 1932, and November 1932, when radio was pro-Weimar, radio availability reduced Nazi vote share. The coefficient on signal strength is about -0.079, and the coefficient on the nonlinear signal measure is about -0.227. In the pooled 1928–1933 panel, signal

TABLE I
RADIO SUBSCRIPTIONS AND SIGNAL STRENGTH

	(1)	(2)	(3)	(4)
	Share of households with radio subscription at a given date			
Date for the subscription rate	Apr 1931	Apr 1932	Apr 1932	Apr 1933
Date for the signal strength	Sep 1930	Jul 1932	Nov 1932	Mar 1933
Panel A				
Radio signal strength	0.218*** [0.030]	0.175*** [0.028]	0.161*** [0.031]	0.163*** [0.036]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R ²	0.576	0.531	0.526	0.514
F-stat for the signal strength variable	50.19	36.13	24.53	20.30
Panel B				
Radio signal strength, nonlinear transformation	0.671*** [0.096]	0.683*** [0.100]	0.656*** [0.120]	0.602*** [0.123]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R ²	0.579	0.533	0.525	0.513
F-stat for the signal strength variable	46.62	42.41	26.38	23.05

Notes. Standard errors clustered by region (Wahlkreise) in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Dependent variable is the number of subscriptions per 100 households. Socioeconomic controls include share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property, share of unemployed and partially employed. Voting controls include turnout and vote shares of DNVP, NSDP, SPD, and Zentrum in December 1924. Number of observations varies because of missing data on listenership and because of redistricting.

TABLE II
DETERMINANTS OF RADIO AVAILABILITY

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Radio signal strength					Radio signal strength, non-linear transformation				
Election date	March 1928	September 1930	July 1932	November 1932	March 1933	March 1928	September 1930	July 1932	November 1932	March 1933
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel A										
Determinants of transmitter location (population, distance to the nearest city with population over 50k, city dummy, altitude)										
Distance to the nearest city, log	-3.187*** [0.650]	-3.531*** [0.655]	-2.147*** [0.598]	-2.589*** [0.598]	-2.862*** [0.482]	-1.949*** [0.221]	-1.188*** [0.233]	-0.819*** [0.166]	-0.538*** [0.140]	-0.812*** [0.140]
Altitude	-0.007* [0.004]	-0.006 [0.004]	-0.009** [0.004]	-0.008** [0.004]	-0.008* [0.004]	-0.002 [0.001]	-0.002 [0.001]	-0.002* [0.001]	-0.002 [0.001]	-0.002** [0.001]
City (Shadkowsky) dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat for determinants of transmitter location	9.184	9.392	10.85	8.271	13.55	9.230	10.28	12.28	11.03	13.88
Panel B										
Socioeconomic controls (shares of Jews and Catholics, shares of blue-collar and white-collar workers, share of war participants, share of welfare recipients, share of renters of social housing, property tax, unemployment, share of unemployed, partial employment)										
Socioeconomic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat for socioeconomic variables	5.008	4.668	2.867	2.453	3.194	5.423	5.509	6.282	3.666	3.495
Panel C										
Voting results in 1924 (turnout and shares of vote received by DNVP, NSDP, SPD, and Zentrum)										
Voting results in 1924	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat for voting in 1924	2.863	3.224	5.981	4.822	5.896	1.358	1.993	3.546	2.039	4.192
R ²	0.668	0.680	0.598	0.609	0.570	0.545	0.555	0.573	0.532	0.489
Observations	959	959	959	959	959	959	959	959	959	959
The share (%) of the total explained variation, explained only by:										
Determinants of transmitter location	38.82	41.18	43.81	37.11	51.93	57.98	62.52	44.85	47.37	57.06
Determinants of transmitter location and Region fixed effects	97.75	97.50	97.16	98.36	97.02	98.90	98.20	98.08	99.62	97.14
Socio-economic controls and voting results in 1924	30.69	30.15	29.26	28.74	34.21	37.61	36.94	30.54	32.89	39.88

interacted with pro-Nazi slant is positive and significant.

Economic message. The sign of radio's effect flips with content. Radio is not mechanically pro-Nazi or anti-Nazi; it amplifies the message of whoever controls the broadcast.

5.7 Table IV: Radio and changes in Nazi vote share

Read. From 1928 to September 1930, stronger radio exposure is associated with smaller Nazi gains. From November 1932 to March 1933, stronger radio exposure is associated with larger Nazi gains. The nonlinear signal coefficient is -0.217 in 1930 and +0.128 in 1933. IV estimates using subscriptions are larger: about -0.347 in 1930 and +0.222 in 1933.

Economic message. Pro-Weimar radio slowed Nazi growth, but the Nazi takeover of radio reversed this effect quickly. This is the clearest evidence that content control is decisive.

5.8 Table V: Referendum and presidential elections

Read. Radio exposure reduced support for the 1929 anti-Versailles Treaty referendum. The signal coefficient is -0.065 and the nonlinear-signal coefficient is -0.244. In the 1932 presidential election, radio exposure is positively associated with votes for incumbent von Hindenburg and negatively associated with votes for Hitler, though not all coefficients are significant.

Economic message. Results beyond Reichstag elections point in the same direction: pro-Weimar radio weakened extremist or anti-system political choices.

	(1)	(2)	(3)	(4)	(5)	(6)
	Nazi vote share		All parliamentary elections 1928-1933, combined		Change in the Nazi vote share	
Time span of the sample	September 1930, July 1932, and November 1932				Between September 1930 and November 1932	
Specification	Panel: OLS	Panel: OLS	Panel: OLS	Panel: OLS	First differences: OLS	First differences: IV
Radio signal strength	-0.079** [0.034]					
Radio signal strength, nonlinear transformation		-0.227** [0.098]				
Radio signal strength $\times$ pro-Nazi slant			0.028*** [0.007]			
Nonlinear transformation of signal strength $\times$ pro-Nazi slant				0.123*** [0.027]		
Change in subscription rate between April 1931 and April 1932					-0.104 [0.098]	-2.127** [0.873]
Baseline controls, interacted with time fixed effects	Yes	Yes	Yes	Yes	No	No
Baseline controls	No	No	No	No	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	No	No
Observations	2,836	2,836	4,713	4,713	827	827
Number of districts	959	959	959	959	827	827
R^2	0.839	0.939	0.972	0.972	0.658	0.658
F-statistics for instrumental variable						26.44

Notes. Standard errors clustered at the region level in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Pro-Nazi slant equals 0 in 1928, -1 between 1930 and 1932, +1 in 1933. Controls include fifth-order polynomial of population, share of Jewish and Catholic population, share of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, altitude, share of unemployed and partially employed, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Number of observations changes between elections because of redistricting. In column (6) change in radio subscription rate is instrumented using change in nonlinear transformation of the signal strength.

TABLE V
RADIO AVAILABILITY AND VOTING IN ANTI-VERSAILLES TREATY REFERENDUM AND APRIL 1932 PRESIDENTIAL ELECTIONS

	(1)	(2)	(5)	(6)	(7)	(8)
	Referendum on the Law against the Enslavement of the German People		1932 presidential elections, 1st round			
	Voted "yes" in the referendum (share of eligible voters)		Von Hindenburg vote share		Hitler vote share	
Panel A: radio availability and other voting outcomes.						
Radio signal strength	-0.065** [0.031]		0.054** [0.026]		-0.049 [0.036]	
Radio signal strength, non-linear transformation		-0.244** [0.095]		0.194 [0.118]		-0.240 [0.149]
Nazi party vote share in 1928	0.647*** [0.090]	0.639*** [0.090]	-0.457*** [0.070]	-0.456*** [0.071]	0.584*** [0.064]	0.534*** [0.064]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R^2	0.845	0.846	0.881	0.881	0.876	0.876
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	-0.138 [0.133]		-0.101 [0.174]		-0.161 [0.157]	
Index of observables (predicted nonlinear transformation of signal strength)		-0.525 [0.393]		-1.291* [0.694]		-0.113 [0.634]
Nazi party vote share in 1928	1.276*** [0.091]	1.293*** [0.090]	-1.490*** [0.129]	-1.509*** [0.119]	1.490*** [0.128]	1.526*** [0.123]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R^2	0.229	0.231	0.202	0.213	0.227	0.225

Notes. Standard errors clustered by region (Wahlkreis) in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Controls include fifth-order polynomial of population, share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, share of unemployed and partially employed, altitude, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924, and Nazi party vote share in 1928.

TABLE IV
RADIO AND AN INCREASE IN NAZI VOTE SHARE

	(1)	(2)	(3)	(4)
	Change in vote share of the Nazi party since previous elections			
Election dates	Sep 1930 (change from May 1928)		Mar 1933 (change from Nov 1932)	
Panel A: reduced-form estimation				
Radio signal strength	-0.061*** [0.022]		0.045** [0.020]	
Radio signal strength, nonlinear transformation		-0.217*** [0.071]		0.128* [0.071]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	958	958	918	918
Panel B: OLS and IV results				
Date for the subscription rate variable	Apr 1931		Apr 1933	
Specification	OLS	IV	OLS	IV
Radio subscription rate, %	-0.086* [0.045]	-0.347*** [0.095]	0.032* [0.017]	0.222* [0.117]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	857	855	853	851
F-statistic for the exclusion of the instrument		50.19		20.30
Panel C: Altonji-Elder-Taber tests				
Index of observables (predicted signal strength)	-0.037 [0.070]		-0.047 [0.049]	
Index of observables (predicted nonlinear signal strength)		0.014 [0.210]		-0.348* [0.200]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Observations	958	958	918	918
R^2	0.432	0.432	0.573	0.576

Notes. Standard errors clustered by region (Wahlkreis) in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Baseline controls include fifth-order polynomial of population, share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, altitude, share of unemployed and partially employed, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Number of observations changes between elections because of redistricting. In Panel B radio subscription rate is instrumented using nonlinear transformation of the signal strength.

TABLE VI
DISCRIMINATION AND VIOLENCE AGAINST JEWS

	(1)	(2)	(3)
	Incidence of discrimination and violence against Jews		
Time span of the sample	1929-1932	1933-1934	1929-1934
Specification	Panel	Pooled cross-section	Panel
Radio signal strength	-0.0020 [0.0019]	0.0020* [0.0011]	
Radio signal strength $\times$ pro-Nazi slant			0.0012** [0.0006]
Baseline controls, interacted with time fixed effects	Yes		Yes
City fixed effects	Yes		Yes
Time fixed effects	Yes	Yes	Yes
Region fixed effects, baseline controls		Yes	
Observations	5,328	2,664	7,992
Number of cities	1332	1332	1332
R^2	0.130	0.338	0.410

Notes. Standard errors clustered by city in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Pro-Nazi slant equals -1 between 1929 and 1932 and +1 between 1933 and 1934. Baseline controls include log(population), altitude, dummy for being located on a navigable river, share of Jewish population in 1925, share of Catholic population in 1925, share of blue-collar workers in 1925, share of white-collar workers in 1925, dummy for the data source, number of war participants per 1,000, number of welfare recipients per 1,000, number of pensioners with social assistance per 1,000, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924.

5.9 Table VI: Discrimination and violence against Jews, 1929–1934

Read. During 1929–1932, radio has a negative but statistically insignificant relationship with anti-Semitic incidents. During 1933–1934, radio has a positive and marginally significant relationship. In the full panel, the interaction of radio signal with pro-Nazi slant is positive and significant.

Economic message. The effect of radio on anti-Semitic behavior also depends on slant. The same medium that had a weak negative relationship with anti-Semitic incidents under pro-Weimar content becomes positively associated with anti-Semitic incidents under Nazi control.

5.10 Table VII: Radio availability and Nazi party membership

Read. Radio exposure has no significant effect on new NSDAP membership in 1932, when radio was pro-Weimar. After the Nazis took over radio, February–May 1933 party membership is positively related to signal strength. The paper estimates that roughly 58% of new party members in the early Nazi-control period were persuaded by radio propaganda.

Economic message. Radio did not just affect voting. It also affected costly political participation in the Nazi Party after the Nazis controlled broadcast content.

Time period	(1)	(2)	(3)	(4)	(5)	(6)
			Log of the number of new party members of NSDAP			
	Jan 1932–Jan 1933		Feb–May 1933			
Panel A: cross-sectional estimates						
Radio signal strength	0.0032 [0.0020]		0.0053** [0.0020]		0.0052** [0.0020]	
Radio signal strength, nonlinear transformation		0.0006 [0.0099]		0.0201** [0.0077]		0.0203** [0.0079]
Log of new party membership in 01/1932–01/1933					0.0544* [0.0292]	0.0579* [0.0289]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R ²	0.677	0.676	0.344	0.345	0.348	0.349
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	0.0026 [0.006]		0.0013 [0.004]		0.0016 [0.004]	
Index of observables (predicted nonlinear signal strength)		0.0221 [0.029]		0.0077 [0.017]		0.0080 [0.017]
Log of new party membership in 01/1932–01/1933					0.0484* [0.027]	0.0480* [0.027]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R ²	0.587	0.587	0.288	0.288	0.292	0.292

Notes. Standard errors clustered by region (Wahlkreis) in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Controls comprise fifth-order polynomial of population, share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, share of unemployed and partially employed, altitude, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924, dummy for pogroms in 1349 and a dummy for a Jewish settlement in 1349. Number of observations changes between elections because of redistricting.

Model	(1)	(2)	(3)
	Log(deportations before 1942) OLS	Letters to <i>Der Stürmer</i> ML	Attacks on synagogues OLS
	All cities	All cities	All cities with synagogues in 1933
Panel A: Baseline results.			
Radio signal strength in 1937	0.0187** [0.0069]	0.0194*** [0.0068]	0.0001 [0.0014]
Log (population)	0.3447*** [0.0803]	0.3900*** [0.0634]	−0.0097 [0.0141]
Altitude	−0.0009* [0.0005]	−0.0008 [0.0005]	−0.0001 [0.0002]
City located at navigable river	0.3693*** [0.0683]	0.7504*** [0.1191]	0.0467** [0.0189]
Socioeconomic controls	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,134
R ²	0.385		0.103
Panel B: Altonji-Elder-Taber style test			
Index of observables (Prediction of signal strength)	−0.0084 [0.0263]	0.0245 [0.0221]	0.0103** [0.0045]
Population	Yes	Yes	Yes
Geographic variables	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,133
R ²	0.197		0.064

Notes. Results of OLS estimation in columns (1) and (3); results of negative binomial maximum likelihood estimation in column (2). Standard errors are clustered by region (Wahlkreis). * $p < .1$, ** $p < .05$, *** $p < .01$. Unit of observation is city in Voigtländer and Voth (2012) sample. Socioeconomic controls include share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property. Voting controls include voter turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Geographic controls include altitude and dummy for being located on a navigable river.

5.11 Table VIII: Radio availability and anti-Semitism after consolidation

Read. Radio signal strength in 1937 significantly increases deportations before 1942 and letters to *Der Stuermer*. The coefficient is about 0.0187 for log deportations and 0.0194 for letters. A one-standard-deviation increase in signal strength implies about 21% more deportations and 22%

TABLE IX
RADIO AVAILABILITY AND ANTI-SEMITISM, THE ROLE OF HISTORIC PREDISPOSITIONS

Model	(1) Log(deportations before 1942) OLS	(2) Letters to <i>Der Stürmer</i> ML	(3) Attacks on synagogues OLS
Panel A: interactions with pogroms			
Sample	Subsample of cities with historical Jewish settlement		
Pogroms in 1349 * Radio signal strength, demeaned	0.048*** [0.016]	0.073*** [0.016]	0.003 [0.007]
Radio signal strength, 1937	-0.032* [0.017]	-0.058*** [0.016]	-0.004 [0.007]
Pogroms in 1349	0.701*** [0.171]	0.601*** [0.227]	0.160** [0.065]
All baseline controls	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	296	319	274
R^2	0.658		0.253
Panel B: interaction with NSFP vote in 1924			
Sample	Full sample of cities		Sample of cities with synagogues
Vote for NSFP in 1924 * Radio signal strength, demeaned	0.131* [0.069]	0.172*** [0.040]	0.014 [0.016]
Radio signal strength, 1937	0.006 [0.007]	0.001 [0.009]	-0.001 [0.002]
Vote for NSFP in 1924	6.003** [2.865]	9.399*** [2.752]	0.822* [0.435]
All baseline controls	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,134
R^2	0.427		0.109
Panel C: interactions with historical inequality			
Sample	Full sample of cities		Sample of cities with synagogues
Land inequality in 1895 * Radio signal strength, demeaned	0.107*** [0.035]	0.094** [0.037]	0.032*** [0.010]
Radio signal strength, 1937	-0.059** [0.025]	-0.052* [0.028]	-0.022*** [0.007]
Land inequality in 1895	1.544 [0.931]	0.173 [1.234]	-0.439** [0.214]
All baseline controls	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Full radio effect for minimal level of inequality (Gini index 45)	-0.010	-0.009	-0.008***
Observations	1,306	1,372	1,116
R^2	0.410		0.117

Notes. Results of OLS estimation in columns (1) and (3); results of negative binomial maximum likelihood estimation in column (2). Standard errors are clustered by region (Wahlkreis). * $p < .1$, ** $p < .05$, *** $p < .01$. Unit of observation is city in Voigtländer and Voth (2012) sample. For Panel A, sample includes only cities with Jewish settlements in 1349. Baseline controls include log(population), altitude, dummy for being located on a navigable river, share of Jewish population in 1925, share of Catholic population in 1925, share of blue-collar workers in 1925, share of white-collar workers in 1925, dummy for Jewish settlement in 1349, number of war participants per 1,000, number of welfare recipients per 1,000, number of pensioners with social assistance per 1,000, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924.

more letters. Effects on synagogue attacks are insignificant.

Economic message. Nazi propaganda affected ordinary citizens' anti-Semitic behavior after dictatorship had consolidated, especially where outcomes were more directly related to citizen expression or denunciation.

5.12 Table IX: Predisposition and propaganda effects

Read. Radio effects are much stronger in places predisposed toward Nazi anti-Semitic messages. Effects are larger where medieval pogroms occurred, where 1924 extreme-nationalist vote share was higher, and where historical land inequality was higher. In cities with historical Jewish settlement but no medieval pogroms, radio exposure reduced deportations and anti-Semitic letters.

Key magnitudes:

- in cities with medieval pogroms, propaganda effects are about 2.6 to 3.8 times larger than in the average city;
- a one-standard-deviation higher NSFP vote share in 1924 roughly doubles the effect on deporta-

tions and strongly raises the effect on letters;

- high land inequality approximately doubles the propaganda effects on deportations and letters;
- in historically less anti-Semitic cities, a one-standard-deviation increase in radio availability is associated with 28% fewer deportations and 45% fewer letters.

Economic message. Propaganda is not uniformly persuasive. It works best when it resonates with prior beliefs and can backfire when it conflicts with those priors.

6 Short summary

Adena et al. study how radio shaped political behavior in Germany before and during Nazi rule. The paper exploits both variation in radio access and sharp changes in radio content. Before 1929, radio was mostly apolitical. From 1929 to early 1933, radio carried pro-Weimar and anti-extremist content. After Hitler became chancellor, the Nazis captured radio and transformed it into a propaganda instrument.

The results show a sharp reversal. Pro-Weimar radio slowed the growth of Nazi electoral support. Nazi-controlled radio increased support for the Nazis in the March 1933 election and increased Nazi party membership. Later, Nazi radio increased anti-Semitic expressions and deportations, especially where listeners were historically predisposed to anti-Semitic propaganda.

The most important conceptual lesson is that media effects depend on content and audience priors. Radio could support democracy when democratic actors controlled the message. The same radio network became dangerous when the Nazis controlled it. Propaganda was powerful, but not universal: it persuaded receptive audiences and sometimes backfired among unreceptive ones.

7 Conclusion

Core conclusion. Mass media can either protect or undermine democracy. In Weimar Germany, radio reduced support for the Nazis when the broadcast was pro-democratic and anti-extremist. After the Nazis captured the state and the radio system, radio increased Nazi support and helped mobilize anti-Semitic behavior.

Economic intuition. Radio is a persuasion technology. Its political effect equals access multiplied by content, filtered through listener predispositions. Access without content control does not have a fixed ideological direction. Content without receptive listeners may fail or backfire.

Identification takeaway. The paper is credible because it does not rely only on cross-sectional differences in radio access. It uses changes in radio content, radio expansion, fixed effects, controls for transmitter placement, placebo tests, IV logic based on signal strength, and historical measures of predisposition.

Broader takeaway. The institutional role of media depends on the political environment. In a fragile democracy, restricting extremist control over mass media can help protect democratic institutions. In dictatorship, propaganda can stabilize the regime, but only when its messages are sufficiently aligned with listeners' prior beliefs.